INCH-POUND

MIL-M-28787/53B(SH)
AMENDMENT 3
14 January 1992
SUPERSEDING
AMENDMENT 2
13 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, DIFFERENTIAL LINE, DIGITAL, KEY CODE KDN

This amendment forms a part of Military Specification MIL-M-28787/53B(EC), dated 4 June 1980, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

* TABLE I, Test conditions L1-L10, Test type column: Delete "X" and substitute "S".

PAGES 4 AND 5

TABLE I, class II column heading: Delete "100°" (two places) and substitute "85°" (two places).

PAGE 6

* 4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through K and M through AA shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3."

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* 4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through K and M through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests D through K and M through AA shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

6.4, second line: Delete "100°C" and substitute "85°C".

PAGE 8

Add the following new paragraph:

"6.7 Cross reference of classification. Military specification MIL-M-28787/53 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658773. Part number M28787/53-1 supersedes part numbers 2658773 and M28787/53."

PAGE 13

FIGURE 6: Delete and substitute new figure 6 as printed on page 3 of this amendment.

PAGE 22

FIGURE 9: Delete and substitute new figure 9 as printed on page 4 of this amendment.

PAGE 23

* FIGURE 10: Delete and substitute new figure 10 as printed on page 5 of this amendment.

PAGE 50

FIGURE 22: Delete and substitute new figure 22 as printed on page 6 of this amendment.

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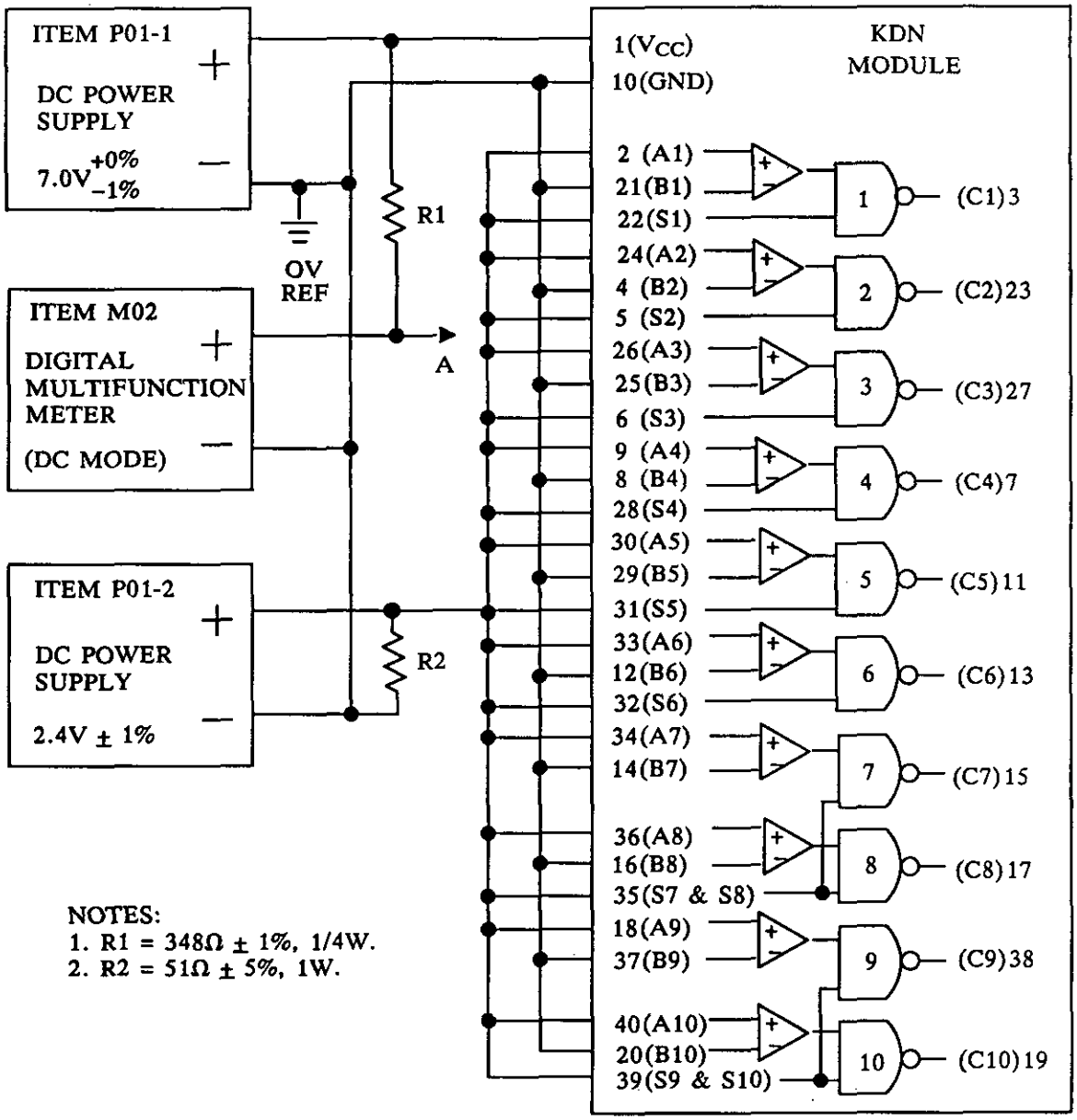
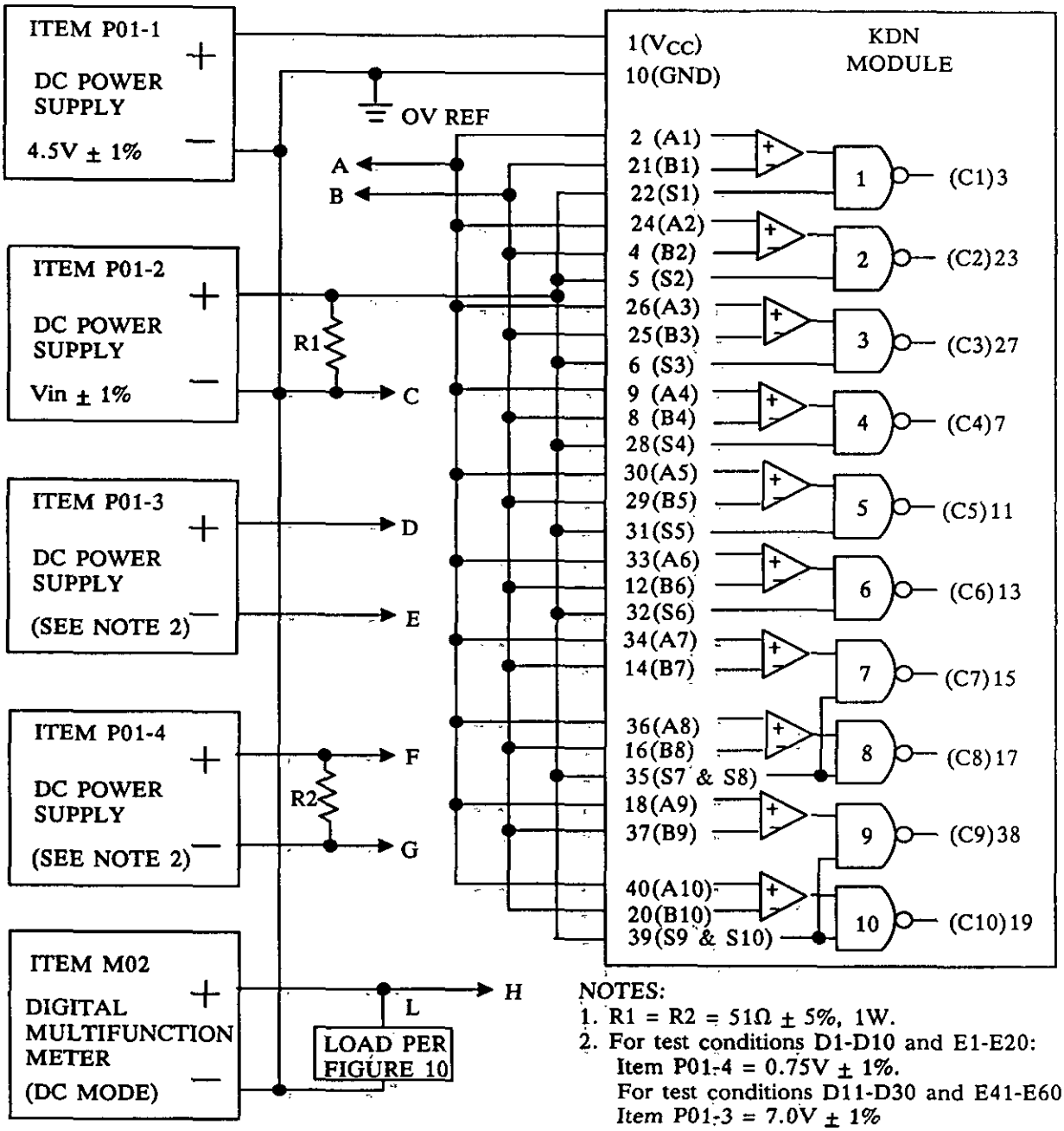


FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

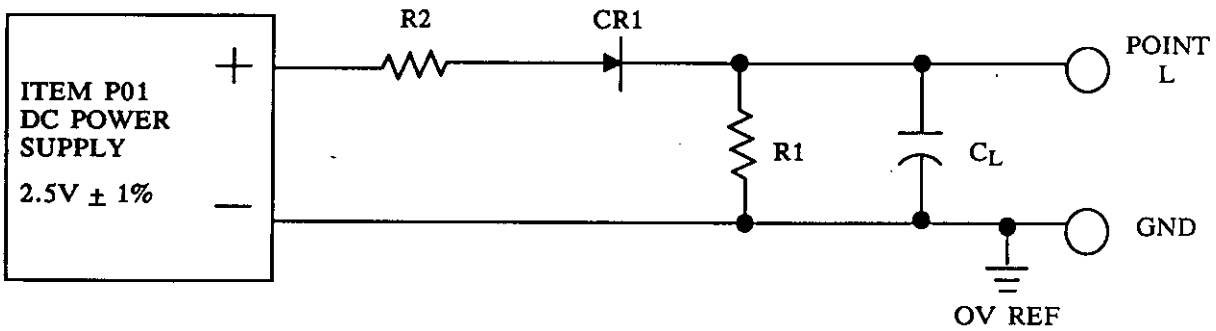
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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1W.
2. For test conditions D1-D10 and E1-E20:
   Item P01-4 = 0.75V ± 1%.
   For test conditions D11-D30 and E41-E60:
   Item P01-3 = 7.0V ± 1%
   Item P01-4 = 0.75V ± 1%.
   For test conditions D31-D50 and E21-E40:
   Item P01-3 = 13.9V ± 1%
   Item P01-4 = 1.0V ± 1%.

FIGURE 9. Static output voltage test circuit for table VI (low-level) and table VII (high-level).

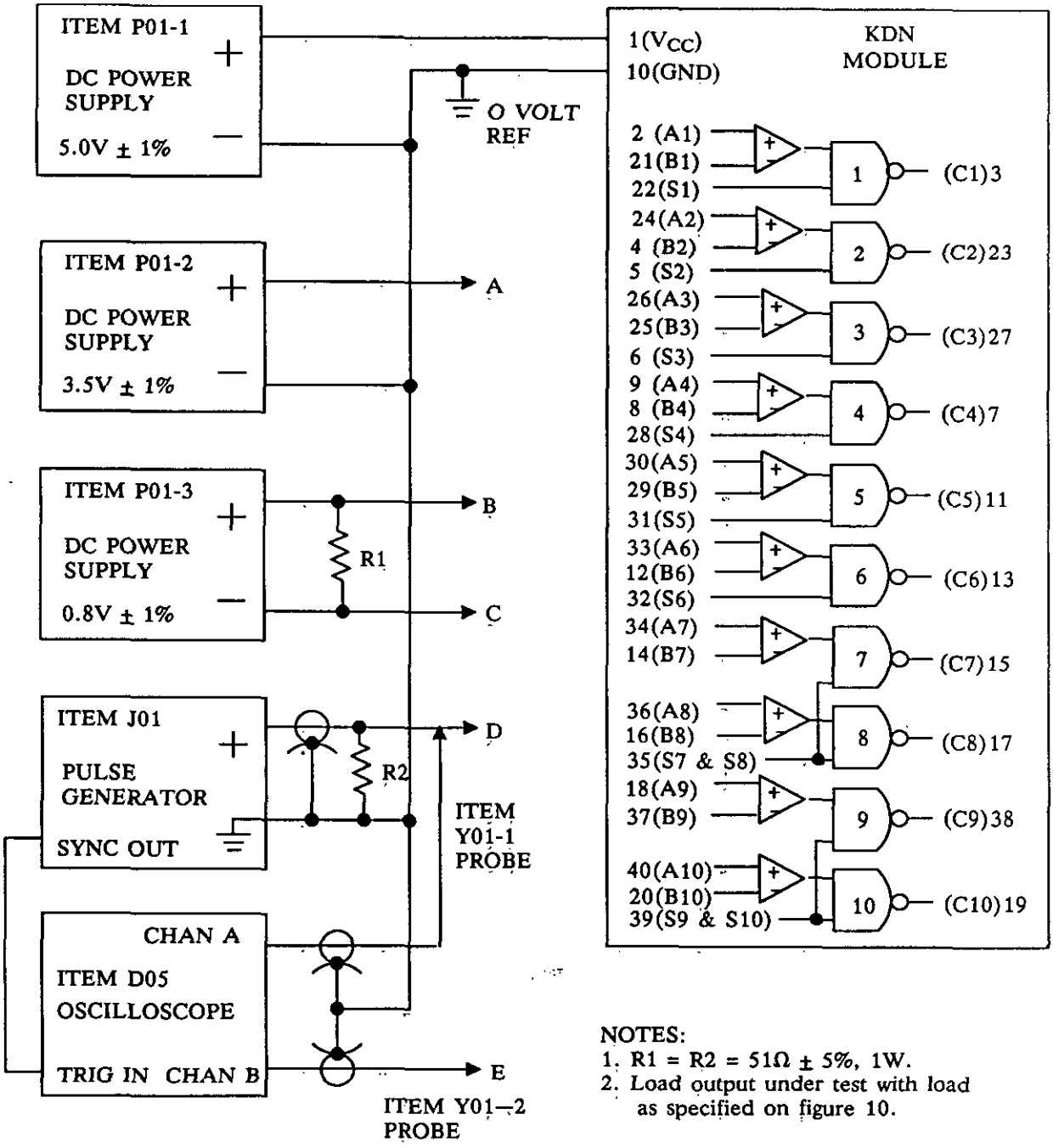
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NOTES:
1. C_L = 100 ± 5pF (including probe and parasitic capacitance) for test conditions N, P, R, S, T, and U.
2. CR1 = 1N4150 or equivalent.
3. R1 = 487Ω ± 1%, for test conditions D and E.
4. R1 = 2.67kΩ ± 1%, for test conditions N, P, R, S, T, U, V, W, Z, and AA.
* 5. R2 = Select value for a current flow of 15mA ± 1% out of point L with point L held at 0.4V.
6. Capacitor C_L is not required for dc electrical test conditions D and E.
7. C_L = 50 ± 5pF (including probe and parasitic capacitance) for test conditions V, W, Z, and AA.
8. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2mm).
9. During output switching, the 2.5V power supply shall have a maximum overshoot of 500mV.

FIGURE 10. Logic load circuit.

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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1W.
2. Load output under test with load as specified on figure 10.

FIGURE 22. Input pulse immunity (high-level and low-level) check test circuit for table XXI.

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